# Different-j full-image setup reuse: result and decision

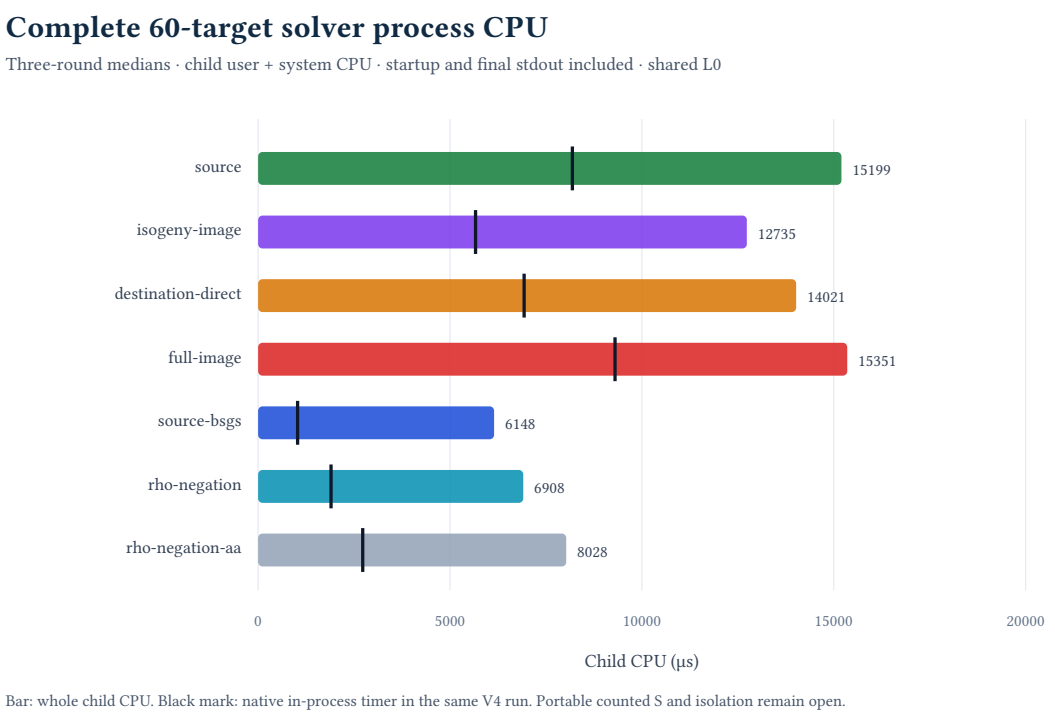
**2026-10-15 · native Rust · Darwin arm64, macOS 26.6, shared L0.** The source and degree-three destination have j=7 and j=11 over $F67^2$. All 60 nonidentity order-61 public targets were solved by each of seven arms in three rotated rounds. Separate native arithmetic replay checked **1,260/1,260** scalar rows and **10,960** relation witnesses in each of V3 and V4.

The V1 target-order formula disclosed answers and was corrected before measurement. The V2 zsh wrapper stopped after one arm and was preserved. V3 completed the native cold-prefix run. V4 repeated the same solver binary and inputs with whole-child CPU, wall and RSS meters. All 21 V4 meter receipts matched raw solver files and replay.

The one table uses **CPU microseconds**, three-round medians. V3 begins before input reading/model construction and includes one setup but excludes process startup and final stdout. V4 charges the complete solver child process at N=60.

| Arm | V3 N=60 native CPU | V4 N=60 process CPU |
|---|---|---|
| Source projected | 5,872.041 | 15,199 |
| Isogeny-image projected | 6,900.125 | 12,735 |
| Destination direct | 5,389.917 | 14,021 |
| Full rational image | 11,049.667 | 15,351 |
| Direct source BSGS | 643.208 | 6,148 |
| Direct negation rho | 2,431.542 | 6,908 |
| Rho A/A | 2,392.959 | 8,028 |

Paired full-image/BSGS ratio medians were **18.103** in V3 N=60 native CPU and **2.483** in V4 complete child CPU. Paired full-image/rho medians were **5.284** and **2.222**. Whole-process overhead changed the ratio, not the algorithm. Full-image native cost per target fell from 4,538.750 µs at N=1 to 184.161 µs at N=60; direct BSGS was 10.720 µs per target at N=60. The 60-target capacity of this subgroup is exhausted.




Bar: whole child CPU. Black mark: native in-process timer in the same V4 run. Portable counted S and isolation remain open.

The full rational-image base changed exact two-sum support membership, yet complete solve cost stayed above both direct controls on this fixed group. Process wall and RSS were recorded, but these are shared-host L0 measurements. Portable counted S, isolated intervals, independent-host cost replay and increasing-order evidence remain open. The source-linked Markdown result and run report retain exact hashes, raw receipts, all prefix/phase ledgers, the V2 failure and the pending larger-order and $\chi$-resultant gates. Canonical frontier charts were left unchanged because no counted-S crossover was admitted.